

Diamonds Do Not Embed Uniformly into Lamplighters on Weighted Binary Trees

Automatic math-discovery packet

July 18, 2026

Source Question

The source paper is Charlotte Melby and Beata Randrianantoanina, *Bi-Lipschitz embedding properties of lamplighter graphs on weighted and unweighted trees*, arXiv:2509.25421. After proving that diamonds do not embed uniformly into lamplighter graphs on the subdivided trees $W_{\theta,n}$ for $\theta \geq 3$, the authors ask whether the same theorem is true for the weighted complete binary trees:

We do not know whether Theorem 1.8 is true for the sequence of lamplighter graphs on the weighted complete binary trees $(\text{La}(\text{WB}_{\theta,n}))_n$, for all or some $\theta \geq 3$.

Next we prove that Theorem 1.5 fails for $(\text{La}(W_{\theta,n}))_n$ when $\theta \geq 3$ in the following much stronger sense (recall that the sequence of diamond graphs $(D_k)_k$ equi-bi-Lipschitzly embeds into the sequence of Hamming cubes $(H_k)_k$, [12], cf. [15]).

Theorem 1.8. *For all $\theta \geq 3$, the sequence of diamonds $(D_k)_{k \in \mathbb{N}}$ does not equi-bi-Lipschitzly embed into the sequence of lamplighter graphs $(\text{La}(W_{\theta,n}))_{n \in \mathbb{N}}$.*

Since $(W_{\theta,n})_n$ are trees, it follows from [6] (cf. [2, Theorem 1.1]) that the sequence $(\text{La}(W_{\theta,n}))_n$ embeds equi-bi-Lipschitzly into ℓ_1 . However, we do not know whether $(\text{La}(W_{\theta,n}))_n$ when $\theta \geq 3$, embed equi-bi-Lipschitzly into every nonsuperreflexive Banach space. If yes, the sequences $(\text{La}(W_{\theta,n}))_n$ for $\theta \geq 3$, would be a new metric characterization of superreflexivity. If no, this would be an even more interesting example.

We do not know whether Theorem 1.8 is true for the sequence of lamplighter graphs on the weighted complete binary trees $(\text{La}(\text{WB}_{\theta,n}))_n$, for all or some $\theta \geq 3$.

Acknowledgements. The second named author is grateful to Chris Gartland for pointing out, during a recent workshop at AIM, the result of Alexandros Eskenazis that finite products of trees do not contain diamonds uniformly (see Corollary 6.2 below), and to the organizers for their invitation to participate in the Workshop on “Metric Embeddings” at the American Institute of Mathematics.

Figure 1: The source paper’s weighted-tree follow-up question, page 5 of the arXiv PDF.

Definitions

Let D_k denote the k -th diamond graph, with graph diameter 2^k . For an integer $\theta \geq 2$, the weighted complete binary tree $\text{WB}_{\theta,n}$ is the depth- n complete binary tree in which every edge

whose lower endpoint has level j has weight θ^{n-j} . Thus the two top edges have weight θ^{n-1} , and each principal subtree below a top edge is isometric to $\text{WB}_{\theta, n-1}$.

For a finite weighted tree T , the lamplighter metric is

$$d_{\text{La}(T)}((A, x), (B, y)) = \text{TSP}_T(x, A \triangle B, y) + |A \triangle B|,$$

where $A, B \subseteq V(T)$, $x, y \in V(T)$, and $\text{TSP}_T(x, S, y)$ is the shortest weighted walk starting at x , ending at y , and visiting every vertex of S .

Proof Intuition

The subdivided trees $W_{\theta, n}$ have many vertices along a long top-scale edge, so the source paper needs a delicate localization argument inside a long path. The weighted trees have no such interior vertices. This makes the top edge a rigid cut.

If two lamplighter states are closer than the weight of a tree edge e , then they cannot be on different sides of e , and they cannot differ by any lamp on the other side of e . Hence a connected graph whose edges all map below that scale is frozen with respect to the cut e . Applied to the two top edges of $\text{WB}_{\theta, n}$, a minimal-depth diamond embedding must have some diamond edge whose image has length at least θ^{n-1} .

But when $\theta \geq 3$, the whole lamplighter graph has diameter at most a constant times θ^n . The image of the two endpoints of D_k then forces $2^k \lesssim D\theta$, where D is the distortion. Thus k is bounded for each fixed D , which rules out equi-bi-Lipschitz embeddings of the entire diamond sequence.

Main Theorem

Theorem 1. *For every integer $\theta \geq 3$, the sequence of diamond graphs $(D_k)_{k \geq 1}$ does not equi-bi-Lipschitzly embed into the sequence $(\text{La}(\text{WB}_{\theta, n}))_{n \geq 1}$.*

Lemma 1 (Diameter bound). *For every $\theta \geq 3$ and $n \geq 1$,*

$$\text{diam } \text{La}(\text{WB}_{\theta, n}) \leq 6\theta^n.$$

Proof. Let $T = \text{WB}_{\theta, n}$. In a tree, a walk visiting any prescribed lamp difference set never needs more than twice the total edge weight of T . Therefore

$$\text{diam } \text{La}(T) \leq 2 \sum_{j=1}^n 2^j \theta^{n-j} + (2^{n+1} - 1).$$

Since $\theta \geq 3$,

$$\sum_{j=1}^n 2^j \theta^{n-j} \leq \theta^n \sum_{j=1}^{\infty} \left(\frac{2}{\theta}\right)^j = \frac{2}{\theta - 2} \theta^n \leq 2\theta^n,$$

and $2^{n+1} - 1 \leq 2\theta^n$. Hence

$$\text{diam } \text{La}(T) \leq 4\theta^n + 2\theta^n = 6\theta^n.$$

□

Lemma 2 (Top-edge freezing). *Let $T = \text{WB}_{\theta, n}$, $n \geq 1$, and let e be one of the two top edges. Removing e separates T into the principal subtree T_e below e and its complement. If two lamplighter states $(A, x), (B, y) \in \text{La}(T)$ satisfy*

$$d_{\text{La}(T)}((A, x), (B, y)) < \theta^{n-1},$$

then x and y lie on the same side of e . Moreover, if they lie outside T_e , then $A \cap T_e = B \cap T_e$, while if they lie inside T_e , then $A \setminus T_e = B \setminus T_e$.

Proof. The edge e has weight θ^{n-1} . Any weighted walk from one side of e to the other must traverse e , so it has length at least θ^{n-1} . Thus the lamplighter positions cannot be on different sides. If the positions are on one side but $A\triangle B$ contains a vertex on the other side, any walk that visits the changed lamp must also traverse e , again giving length at least θ^{n-1} . This proves the claim. $\square$

Lemma 3 (Minimal-depth lower scale). *Fix $D \geq 1$, $k \geq 1$, and $\theta \geq 3$. Suppose n is minimal among all depths for which there is a D -bi-Lipschitz embedding $\phi : D_k \rightarrow \text{La}(\text{WB}_{\theta,n})$. If $\lambda > 0$ is a scaling factor for ϕ , so that*

$$\lambda d_D(u, v) \leq d_{\text{La}(\text{WB}_{\theta,n})}(\phi(u), \phi(v)) \leq D\lambda d_D(u, v),$$

then

$$D\lambda \geq \theta^{n-1}.$$

Proof. Assume instead that $D\lambda < \theta^{n-1}$. Since every edge of D_k has length 1, adjacent vertices of D_k map to lamplighter states at distance $< \theta^{n-1}$.

Apply Lemma 2 to each of the two top edges of $\text{WB}_{\theta,n}$. Because D_k is connected, for each top edge the side containing the lamplighter position is constant throughout $\phi(D_k)$.

If all lamplighter positions lie in one principal subtree T_e , then Lemma 2 also says that the lamp configuration outside T_e is constant over the image. Toggling this fixed outside set is an isometry of the lamplighter graph, since symmetric differences are unchanged. After this isometry the image lies in $\text{La}(T_e)$. But T_e is isometric to $\text{WB}_{\theta,n-1}$, contradicting the minimality of n .

The only remaining possibility is that all lamplighter positions lie outside both principal subtrees, hence at the root. Then the lamp configuration in each principal subtree is constant over the image. After toggling those two fixed lamp sets, only the root lamp can vary, so the image has at most two points. This is impossible for an injective bi-Lipschitz image of D_k with $k \geq 1$.

Thus $D\lambda \geq \theta^{n-1}$. $\square$

Proof of Theorem 1. Suppose, toward a contradiction, that the sequence (D_k) embeds equi-bi-Lipschitzly into $(\text{La}(\text{WB}_{\theta,n}))$ with distortion at most D . For each k , choose n minimal among all target depths admitting such an embedding, and let λ be its scaling factor.

Let s, t be the two original endpoints of D_k , so $d_D(s, t) = 2^k$. By Lemma 1,

$$\lambda 2^k \leq d_{\text{La}(\text{WB}_{\theta,n})}(\phi(s), \phi(t)) \leq 6\theta^n.$$

By Lemma 3, $\theta^{n-1} \leq D\lambda$. Combining the two estimates gives

$$2^k \leq \frac{6D\theta^n}{\theta^{n-1}} = 6D\theta.$$

For fixed D and θ , this bounds k , contradicting the existence of uniformly distorted embeddings for all diamond levels. $\square$

Verification Notes

The accompanying script `code/verify_scale_bounds.py` checks the elementary finite-sum bound

$$2 \sum_{j=1}^n 2^j \theta^{n-j} + (2^{n+1} - 1) \leq 6\theta^n$$

for a finite range of θ, n , and checks the resulting threshold $2^k > 6D\theta$. The proof above gives the general symbolic argument.

Limitations and Novelty Check

This packet answers only the second follow-up question on page 5 of the source paper: the no-diamonds theorem for weighted complete binary trees. It does not answer the separate question whether $(\text{La}(W_{\theta,n}))_n$ for $\theta \geq 3$ embeds into every nonsuperreflexive Banach space.

The run indexes were searched for arXiv:2509.25421, lamplighter graphs, weighted binary trees, diamonds, and the adjacent BMSZ Problem 7.1 family. The only relevant prior attempt was an older stretched-tree push for arXiv:1902.07098; that target is answered by the present source paper and is not duplicated here. A bounded web search on July 18, 2026 for exact phrases involving “Lam(WB)”, “weighted complete binary trees”, lamplighter graphs, and diamonds found the source paper but no later answer to the weighted-tree follow-up.

References

- C. Melby and B. Randrianantoanina, *Bi-Lipschitz embedding properties of lamplighter graphs on weighted and unweighted trees*, arXiv:2509.25421.
- F. Baudier, P. Motakis, T. Schlumprecht, and A. Zsak, *On the bi-Lipschitz geometry of lamplighter graphs*, Discrete Comput. Geom. 66 (2021), 203–235.
- W. B. Johnson and G. Schechtman, *Diamond graphs and super-reflexivity*, J. Topol. Anal. 1 (2009), 177–189.

Human Review Recommendation

Check the cut-freezing Lemma 2 against the source paper’s normalization of weighted lamplighter metrics. Under the stated TSP-plus-lamp metric, the proof is insensitive to all constants except the strict comparison $D\lambda < \theta^{n-1}$, and equality is handled by the contrapositive in Lemma 3.